Supplementary Material

S1: Statistical Analysis and Reporting of Experimental Results

In the AVEC 2014 dataset, we adopted the same fixed-length temporal normalization strategy as that used in the EIVE dataset. Before hypergraph construction and STHC feature extraction, each complete FAU sequence was linearly resampled along the temporal dimension and uniformly mapped to 60 time points. This process preserves the overall dynamic patterns of the complete sequence without direct truncation or zero-padding, thereby avoiding additional biases introduced by varying sequence lengths or padding boundaries. It is worth noting that the average sequence length of the original FAU recordings in AVEC 2014 is approximately 54 seconds. This setting not only maintains consistency with the sequence length used in the EIVE dataset but also closely matches the intrinsic temporal duration of AVEC 2014.

After temporal normalization, all samples share an identical temporal dimension. The hyperedge statistical features, including the mean, standard deviation, minimum, and maximum values, are calculated based on the same number of temporal sampling points. Therefore, variations in the original video duration do not directly affect the dimensionality of the hyperedge features or the input length of STHC, thereby improving the comparability of temporal representations across different subjects. We further investigated the impact of different temporal sequence lengths on model performance by setting the normalized FAU sequences to 30, 60, and 90 temporal points, respectively. The results are presented in **Tab.1**. The experimental results demonstrate that DEP-FAUs exhibits good robustness to the temporal length of normalized sequences. Compared with 30 temporal points, increasing the sequence length to 60 temporal points improved the ACC and F1 scores of the Northwind task from 0.700 and 0.700 to 0.720 and 0.730, respectively. Similarly, for the Freeform task, ACC and F1 increased from 0.600 and 0.610 to 0.620 and 0.630, respectively. When the temporal length was further increased to 90 points, the overall performance remained largely consistent with that achieved using 60 points, with only a slight decrease in the F1 score of the Northwind task to 0.720. These results indicate that the model performance remains stable across different temporal sequence lengths.

Tab.1. Effect of different temporal lengths on model performance in cross-dataset evaluation.

AVEC 2014/AUS	Northwind		Freeform	
Temporal length	ACC	F1	ACC	F1
30	<u>0.700</u>	0.700	<u>0.600</u>	<u>0.610</u>
60	**0.720**	**0.730**	**0.620**	**0.630**
90	**0.720**	<u>0.720</u>	**0.620**	**0.630**

It should be noted that variations in the original video duration may cause linear resampling to produce inconsistent temporal scales between adjacent points in the normalized sequences and introduce frequency-domain distortions, potentially affecting local temporal features and hyperedge statistical representations. Although this approach standardizes sequence lengths while preserving overall temporal trends, it remains limited in capturing short-term dynamics.

Furthermore, EIVE and AVEC 2014 differ in their tasks and data collection protocols. EIVE employs standardized emotional stimulation tasks with relatively consistent video durations, whereas AVEC 2014 primarily captures facial behaviors during natural interactions, resulting in

substantial variations in video length. Consequently, these temporal distortions may be more pronounced when processing variable-length sequences in AVEC 2014, while their potential impact on EIVE is relatively limited due to its standardized acquisition protocol.

S2: Statistical Power and Sample Size Sensitivity Analysis Based on Raw FAU Effect Sizes

Building on the results presented in Table X of the manuscript, we further conducted a statistical power analysis based on the effect sizes of the raw FAU features, along with a sample size sensitivity analysis, to provide a more comprehensive assessment of the statistical characteristics of the EIVE dataset.

First, we conducted a statistical power analysis based on raw FAU features. Following the effect size calculation strategy described above, we aggregated each subject's raw FAU time series along the temporal dimension to obtain independent subject-level FAU representations. Cohen's d was then used to quantify the standardized differences between the depression and healthy control groups. Given that the EIVE dataset comprises 80 patients with depression and 80 healthy controls, we employed a two-sided independent-samples t-test framework with a significance level of 0.05 and calculated the statistical power corresponding to different FAU effect sizes using the noncentral t-distribution. Analyses were performed on raw FAU features prior to model training, without involving model predictions.

The experimental results are presented in **Tab. 2**. The conditional statistical power of seven representative FAUs across the three emotional stimulus conditions ranged from 0.2782 to 0.7655. Among them, AU_17_r under positive emotional stimulation exhibited the largest standardized effect size (|d| = 0.427), corresponding to a statistical power of 0.7655. Overall, the current sample size provides reasonable power to detect relatively pronounced between-group differences in FAUs, although its ability to detect smaller effects remains limited.

Tab. 2. Statistical Power Analysis Based on Raw FAU Effect Sizes.

FAUs	Positive |d|	Power	Negative |d|	Power	Neutral |d|	Power
AU_17_r	0.427	0.7655	0.319	0.5181	0.370	0.6428
AU_07_r	0.359	0.6167	0.291	0.4481	0.312	0.5006
AU_06_r	0.351	0.5974	0.288	0.4406	0.296	0.4605
AU_04_r	0.336	0.5605	0.284	0.4307	0.312	0.5006
AU_26_c	0.230	0.3039	0.326	0.5356	0.351	0.5974
AU_45_r	0.289	0.4431	0.312	0.5006	0.279	0.4184
AU_10_c	0.328	0.5406	0.286	0.4357	0.218	0.2782

Second, to further assess the statistical detection capability under a fixed sample size, we conducted a sample size sensitivity analysis. Specifically, with a total sample size of N = 160, balanced group allocation, a two-sided significance level of 0.05, and a target statistical power of 0.80, we determined the minimum detectable standardized effect size. The results indicate that the current sample configuration can detect a true standardized between-group difference of |d| = 0.446 with 80% statistical power. This finding further clarifies the statistical detection limits of the EIVE dataset for univariate FAU comparisons.

S3: Analysis of Model Robustness, Statistical Significance, and Clinical Severity Association

In this section, we further investigate the following aspects to validate the reliability and clinical applicability of DEP-FAUs:

a. How stable are the reported results across different random splits?

b. What are the confidence intervals or statistical variations of the reported metrics?

c. Is the observed performance improvement statistically significant?

d. How does the model performance vary according to depression severity?

Regarding concern **(a)**, namely that different random data splits may lead to fluctuations in model performance, we further evaluated the stability of DEP-FAUs using repeated five-fold cross-validation with 10 different random seeds (10×5-fold cross-validation). Specifically, in each repeated experiment, DEP-FAUs and all comparison methods were evaluated using exactly the same data partitions to ensure a fair comparison. The mean and standard deviation of model performance were then calculated based on the results from the 10 repeated experiments. As shown in **Tab. 3**, DEP-FAUs maintained stable performance across different random splits. For example, under the Positive, Neutral, and Negative emotional conditions, DEP-FAUs achieved ACC values of 0.756±0.021, 0.693±0.027, and 0.733±0.017, respectively, and F1 scores of 0.763±0.014, 0.696±0.021, and 0.729±0.026, respectively. Compared with the original five-fold cross-validation results, only minor performance variations were observed across different random partitions, indicating that the performance advantages of DEP-FAUs are not dependent on any specific data split.

Regarding concern **(b)**, namely the uncertainty of model performance, we further reported the standard deviation and 95% confidence intervals (CIs) of DEP-FAUs based on the original five-fold cross-validation results. As shown in **Tab.2**, DEP-FAUs achieved ACC values of 0.756±0.015, 0.700±0.017, and 0.731±0.017 under the Positive, Neutral, and Negative conditions, respectively. The corresponding 95% CIs were [0.7374, 0.7746], [0.6787, 0.7213], and [0.7101, 0.7525], respectively. In addition, the corresponding confidence intervals for other evaluation metrics, including F1 score, Recall, and Precision, were also reported. These results demonstrate that the performance variations of DEP-FAUs across different testing folds remained within a reasonable range, further validating the reliability of the model predictions.

Regarding concern **(c)**, to determine whether the performance improvements of DEP-FAUs over existing methods were statistically significant, we conducted paired statistical analyses based on the results of 10 repeated five-fold cross-validation experiments. As shown in **Tab.3**, under each emotional condition, the best-performing representative baseline model was selected (ModernTCN for the Positive condition, and CrossFormer for the Neutral and Negative conditions). The performance differences between DEP-FAUs and the corresponding baseline models were evaluated using the paired

Wilcoxon signed-rank test under identical random data splits, with the Holm method applied for multiple comparison correction. The results showed that, under the Positive emotional stimulus condition, DEP-FAUs achieved statistically significant improvements over ModernTCN in both ACC and F1 metrics (Holm-adjusted $p < 0.05$). Similarly, under the Negative emotional stimulus condition, DEP-FAUs demonstrated significant performance advantages compared with CrossFormer. For the Neutral emotional stimulus condition, the performance improvement did not reach statistical significance after multiple comparison correction, likely because facial behavioral differences between patients and controls are relatively weaker under neutral stimuli, resulting in a limited improvement margin. Nevertheless, DEP-FAUs still maintained a stable performance advantage. These results indicate that the improvements achieved by DEP-FAUs are not attributable to a single random data split but instead demonstrate robust statistical stability across different data partitions.

Regarding concern **(d)**, to further investigate the relationship between DEP-FAUs and clinical depression severity, we conducted a subgroup analysis of 80 patients with depression based on their HAMD scores. According to clinical severity, the patients were categorized into three subgroups: subthreshold or mild depression (HAMD score: 8–20), moderate depression (HAMD score: 20–35), and severe depression (HAMD score >35). Since DEP-FAUs models FAU dynamic patterns separately under different emotional stimuli, the severity analysis was performed at the subject level. Specifically, for each patient with depression, the FAU sequences under the Positive, Neutral, and Negative emotional stimulus conditions were individually input into the model to obtain the corresponding depression prediction probabilities. The prediction probabilities from the three emotional conditions were then averaged for each subject to generate the final depression risk score, which was subsequently used for severity stratification based on HAMD scores. The results are presented in **Tab.4**. As the HAMD score increased, both the model discrimination ability and prediction confidence showed a gradual increasing trend. The sensitivities for patients with mild, moderate, and severe depression were 0.667, 0.778, and 0.824, respectively, with corresponding mean prediction probabilities of 0.624±0.176, 0.731±0.151, and 0.816±0.128, respectively. Furthermore, a significant positive correlation was observed between HAMD scores and model prediction probabilities (Spearman $r = 0.42$, $p < 0.001$), indicating that DEP-FAUs not only enables depression state classification but also captures facial behavioral patterns associated with clinical symptom severity.

Tab.2. Comparative experimental performance of different models on the EIVE dataset under various emotional stimuli. The symbol '_' denotes the second-best performing model.

EIVE/AUS	**Positive**				**Neutral**				**Negative**			
Model	**Acc**	**F1**	**Rec**	**Pre**	**Acc**	**F1**	**Rec**	**Pre**	**Acc**	**F1**	**Rec**	**Pre**
Original FAUs data												

Informer	0.731± 0.041	0.722±0.044	0.700±0.048	**0.770**±0.045	0.643±0.047	0.605±0.048	0.575±0.042	**0.719**±0.050	0.687±0.054	0.656±0.057	0.662±0.041	0.695±0.052
PatchTST	0.662± 0.058	0.696±0.051	0.787±0.068	0.648±0.060	0.606±0.035	0.544±0.054	0.537±0.021	0.648±0.046	0.637±0.055	0.634±0.051	0.737±0.047	0.694±0.043
CrossFormer	0.706±0.040	0.659± 0.067	0.625±0.052	0.730±0.044	0.693±0.034	0.698±0.045	**0.725**±0.051	0.682±0.045	0.718±0.051	0.719±0.055	0.725±0.094	0.722±0.059
Mamba	0.693±0.049	0.680±0.053	0.662±0.046	0.705±0.066	0.656±0.046	0.608±0.059	0.575±0.055	0.710±0.061	0.675±0.055	0.644±0.057	0.650±0.047	0.697±0.061
ModernTCN	0.737±0.052	0.741±0.041	0.750±0.068	0.738±0.059	0.668±0.035	0.676±0.047	0.712±0.055	0.675±0.024	0.700±0.060	0.710±0.058	**0.762**±0.046	0.681±0.057
TimeKAN	0.637±0.051	0.610±0.047	0.600±0.051	0.656±0.050	0.575±0.040	0.607±0.061	0.687±0.069	0.573±0.068	0.612±0.038	0.600±0.054	0.625±0.063	0.620±0.032
DLinear	0.675±0.047	0.654±0.037	0.612±0.052	0.712±0.050	0.643±0.017	0.635±0.028	0.625±0.056	0.655±0.066	0.643±0.068	0.622±0.055	0.587±0.061	0.661±0.057
TimesNet	0.718±0.035	0.713±0.043	0.700±0.058	0.751±0.067	0.656±0.026	0.666±0.048	0.700±0.018	0.652±0.040	0.631±0.047	0.543±0.042	0.537±0.041	0.737±0.052
Hypergraph data												
STGCN	0.706±0.029	0.737±0.037	0.825±0.041	0.667±0.039	0.649±0.037	0.626±0.043	0.587±0.048	0.672±0.051	0.681±0.038	0.634±0.044	0.575±0.041	**0.751**±0.039
HyperGCN	0.712±0.031	0.747±0.039	**0.850**±0.041	0.668±0.045	0.643±0.042	0.617±0.038	0.575±0.049	0.667±0.041	0.668±0.051	0.662±0.053	0.650±0.047	0.675±0.045
Mgrr-net	0.700±0.028	0.727±0.033	0.800±0.047	0.666±0.041	0.631±0.042	0.619±0.051	0.600±0.051	0.642±0.048	0.662±0.045	0.653±0.051	0.637±0.055	0.670±0.047
Dhyper	0.718±0.039	0.742±0.047	0.815±0.042	0.684±0.046	0.662±0.029	0.651±0.037	0.637±0.039	0.671±0.035	0.693±0.053	0.701±0.049	0.725±0.044	0.682±0.039
GraphEx	0.725±0.041	0.749±0.037	0.825±0.033	0.687±0.046	0.668±0.048	0.656±0.039	0.637±0.052	0.681±0.041	0.674±0.039	0.673±0.047	0.675±0.046	0.674±0.036
DEP-FAUs	**0.756**±0.015	**0.761**±0.018	0.787±0.025	0.746±0.022	**0.700**±0.017	**0.704**±0.020	**0.725**±0.026	0.701±0.058	**0.731**±0.017	**0.726**±0.056	0.737±0.032	0.744±0.088
DEP-FAUs (CI)	[0.7374, 0.7746]	[0.7387, 0.7833]	[0.7560, 0.8180]	[0.7187, 0.7733]	[0.6787, 0.7213]	[0.6780, 0.7300]	[0.6924, 0.7576]	[0.6154, 0.7603]	[0.7101, 0.7525]	[0.6565, 0.7963]	[0.6969, 0.7781]	[0.6087, 0.8281]

Tab.3. Model stability and statistical significance analysis based on **10 repeated five-fold cross-validation**.

Emotion	Metric	DEP-FAUs	95% CI	Baseline	95% CI	Wilcoxon pp	Holm pp
Positive	ACC	0.756 ± 0.021	[0.7410, 0.7710]	0.734 ± 0.044 **ModernTCN**	[0.7025, 0.7655]	0.0039	0.0195
	F1	0.763 ± 0.014	[0.7530, 0.7730]	0.735 ± 0.035 **ModernTCN**	[0.7100, 0.7600]	0.0020	0.0117
Neutral	ACC	0.693 ± 0.027	[0.6737, 0.7123]	0.690 ± 0.020 **CrossFormer**	[0.6757, 0.7043]	0.0742	0.1484
	F1	0.696 ± 0.021	[0.6810, 0.7110]	0.687 ± 0.029 **CrossFormer**	[0.6663, 0.7077]	0.0469	0.0938
Negative	ACC	0.733 ± 0.017	[0.7208, 0.7452]	0.731 ± 0.031 **CrossFormer**	[0.7088, 0.7532]	0.0195	0.0390
	F1	0.729 ± 0.026	[0.7104, 0.7476]	0.724 ± 0.029 **CrossFormer**	[0.7033, 0.7447]	0.0098	0.0293

Tab.4. Model performance analysis across different depression severity levels stratified by HAMD scores.

HAMD	n	Sensitivity	Mean prediction probability
8-20	18	0.667	0.624±0.176
20-35	45	0.778	0.731±0.151
>35	17	0.824	0.816±0.128